Grade: Check Plus

Assignment 2

Price (x)	Probability f(x)	x*f(x)	$(x-\mu)^2 * f(x)$
20	0.05	1	1.463405
21	0.06	1.26	1.166886
22	0.07	1.54	0.813967
23	0.08	1.84	0.464648
24	0.09	2.16	0.178929
25	0.12	3	0.020172
26	0.15	3.9	0.052215
27	0.13	3.51	0.328653
28	0.1	2.8	0.67081
29	0.1	2.9	1.28881
30	0.05	1.5	1.053405
		25.41	7.5019

✓ Q1

1 μ = 25.41
standard deviation = 2.74

2a P (25.41 - 2.74 ≤ A ≤ 25.41 + 2.74) =
P (22.67 ≤ A ≤ 28.15) = ✓ Q2a
P (23 ≤ A ≤ 28) =
f(23) + f(24) + f(25) + f(26) + f(27) + f(28) ... OR ... 1 - (f(30) + f(29) + f(22) + f(21) + f(20))
.67 or 67%

2b P (25.41 - 2.74*2 ≤ A ≤ 25.41 + 2.74*2) =
P (25.41 - 5.48 ≤ A ≤ 25.41 + 5.48) = ✓ Q2b
P (19.93 ≤ A ≤ 30.89) =
f(20) + f(21) + f(22) + f(23) + f(24) + f(25) + f(26) + f(27) + f(28) + f(29) + f(30) =
1 or 100%

3

Option value v	Probability P(v)
0	0.47
1	0.15
2	0.13
3	0.1
4	0.1
5	0.05

✓ Q3

✓ Q4

4 f(26) + f(27) + f(28) + f(29) + f(30) =
.53 or 53%

5

Option value v	Probability P(v)	x*P(x)	$(x-\mu)^2 * P(x)$
0	0.47	0	0.869312
1	0.15	0.15	0.01944
2	0.13	0.26	0.053248
3	0.1	0.3	0.26896
4	0.1	0.4	0.69696
5	0.05	0.25	0.66248
		1.36	2.5704

**.47 is derived from adding P(x) for f(20) through f(25)*

✓ Q5 1.603246706

μ = 1.36

standard deviation = 1.6

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